Enhancing Medical Information Extraction via rsLoRA: Fine-Tuning Qwen3-8B for ADE and DDI Recognition

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COMP6713 – Advanced Natural Language Processing

Background & Clinical Significance

- **The Data Overload:** Millions of medical articles exist; manually finding drug safety data is no longer possible.
- **The Safety Risk:** Missing Adverse Drug Events (ADE) and Drug-Drug Interactions (DDI) can lead to severe patient harm.
- **Our Goal:** Build an automated AI tool to accurately extract these hidden medical relationships from complex texts.

Problem Definition & Key Challenges

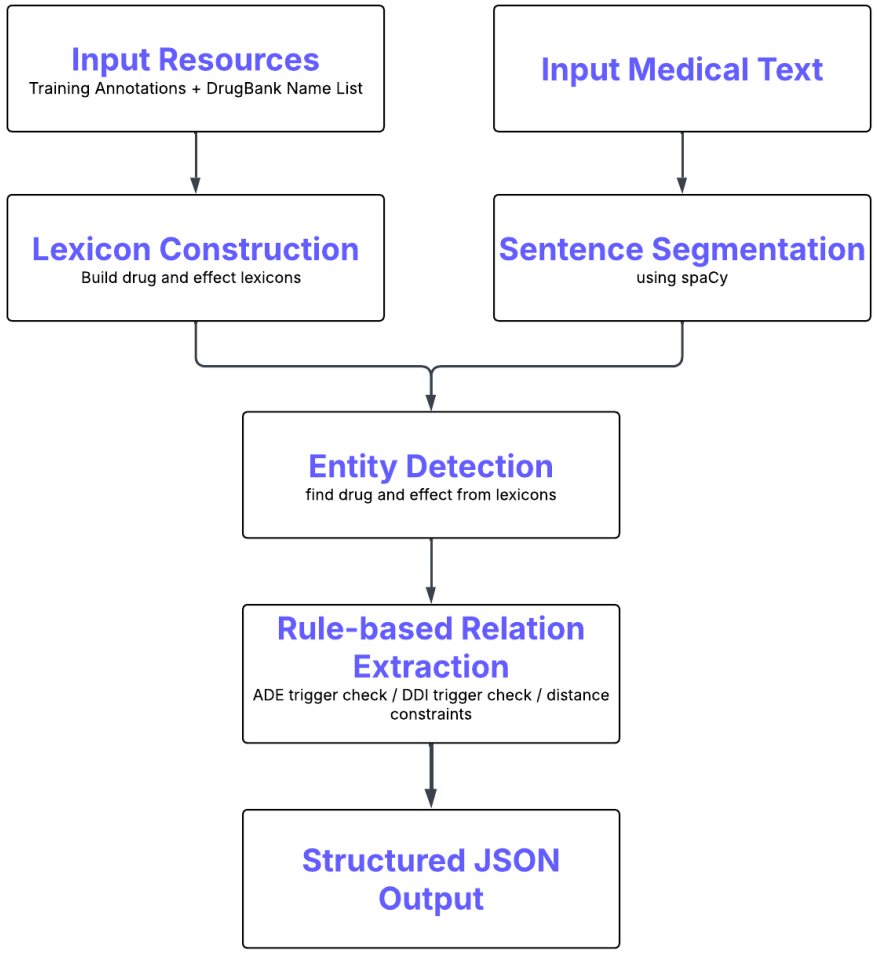
- **The Task:** Automatically read medical texts, spot the drugs/symptoms, and determine exactly how they are linked.
- **Challenge 1: Complex Language:**
  - Medical sentences are often long, twisted, and hide relationships across multiple clauses.
- **Challenge 2: Blurry Boundaries:**
  - It's hard for models to tell the difference between a chemical "mechanism" and a doctor's "advice".
- **Challenge 3: Data Shortage:**
  - High-quality, human-labeled medical data is rare, expensive, and heavily unbalanced.

Data Engineering: Alignment & Augmentation

- **Core Datasets:** Integrated **ADE Corpus V2** and **DDI Extraction 2013** into a unified pipeline.
- **Schema-Constrained Generation:** Employs ChatML to force strictly serialized JSON outputs, ensuring structured data and zero hallucinations.
- **Smart Data Augmentation:** Utilized a generative LLM as a paraphraser to safely expand the DDI training set to 2,000 high-fidelity instances.
- **Global Shuffle Strategy:** Randomized task distribution across both datasets to prevent bias and catastrophic forgetting.

Baseline Systems: The Performance Floor

- A lightweight lexicon & rule baseline
- Drug and adverse-effect lexicons are constructed from annotated training relations.
- The baseline applies sentence-level rules:
  1. ADE: link an effect mention to the nearest drug mention when ADE trigger words appear.
  2. DDI: link nearby drug pairs when DDI trigger words appear.
- Validation tuning is used to adjust simple rule parameters, such as token distance and trigger coverage.



Zero-Shot

- Zero-shot setting tests whether a general LLM can perform ADE/DDI extraction without task-specific fine-tuning.

- We provide task instructions and require structured JSON output.

- The same parser and evaluator are used for zero-shot, fine-tuned models, and baseline outputs.

- Key challenges:
  1. inconsistent JSON formatting
  2. label variations
  3. hallucinated or missing relations

Methodology: Model Selection & Rationale

	Llama-3-8B Base	Qwen2.5-7B Base	Qwen2.5-14B Base	Qwen3-8B Base
Architecture	Dense	Dense	Dense	Dense
# Total Params	8B	7B	14B	8B
# Activated Params	8B	7B	14B	8B
General Tasks				
MMLU	66.60	74.16	**79.66**	76.89
MMLU-Redux	61.59	71.06	**76.64**	76.17
MMLU-Pro	35.36	45.00	51.16	**56.73**
SuperGPQA	20.54	26.34	30.68	**31.64**
BBH	57.70	70.40	78.18	**78.40**
Math & STEM Tasks				
GPQA	25.80	36.36	32.83	**44.44**
GSM8K	55.30	85.36	**90.22**	89.84
MATH	20.50	49.80	55.64	**60.80**
Coding Tasks				
EvalPlus	44.13	62.18	60.70	**67.65**
MultiPL-E	31.45	50.73	54.79	**58.75**
MBPP	48.40	63.40	69.00	**69.80**
CRUX-O	36.80	48.50	61.10	**62.00**
Multilingual Tasks				
MGSM	38.92	63.60	74.68	**76.02**
MMMLU	59.65	71.34	**78.34**	75.72
IINCLUDE	44.94	53.98	**60.26**	59.40

- **Base model: Qwen3-8B**
- Balances **performance** and **cost**

Methodology: rsLoRA Fine-tuning

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Method	Backbone storage	Scaling	Main role in this project
LoRA	Frozen full precision	α/r	Standard PEFT baseline with low-rank trainable updates
QLoRA	Frozen 4-bit quantized backbone	LoRA-style scaling on quantized weights	Lower memory usage when hardware is the main bottleneck
rsLoRA	Frozen full precision or quantized backbone	$\alpha/\sqrt{r}$	More stable scaling when the rank is not too small

- **Fine-tuning method: rsLoRA**
- More stable scaling than standard LoRA at higher ranks
- Preferred over QLoRA

Data Balancing & Training Setup

Loss and reweighting strategy

Weighted sampling

- Empty samples
- DDI-bearing
- DDI-INT
- Multi-relation outputs

$$p(i) = \frac{\omega_i^{(s)}}{\sum_{k=}^{N} \omega_k^{(s)}},$$

Weighted loss

- Label smoothing: $\varepsilon = 0.02$
- Loss weights $\omega_k^{(t)} \in \{1.0, 1.35, 2.4, 1.1\}$

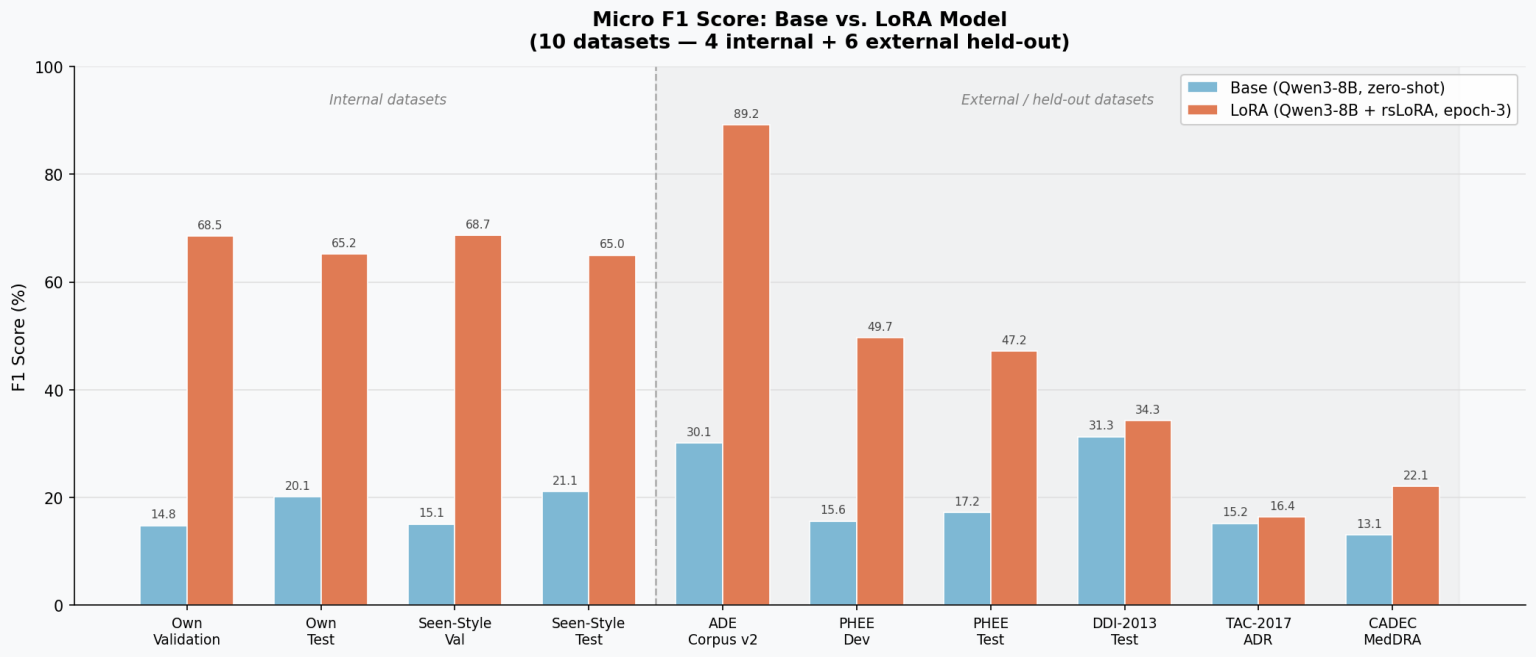
Training setup

Item	Details
Model	Qwen3-8B + rsLoRA
GPU	NVIDIA RTX 6000
Epochs	3
Steps	993
Imbalance handling	Weighted sampling + Weighted loss

Trained Qwen3-8B + rsLoRA over imbalanced data on a single RTX 6000 GPU.

- **Datasets Balance and training setting**

Quantitative Evaluation: Performance Gains



Summary

Internal Avg F1

Rule 10% → Base 22% → LoRA 65%

External Avg F1

Base 15% → LoRA 34%

Best External

ADE Corpus v2: 89.2% F1

+49.1%

Internal F1 Gain (avg)

+22.7%

External F1 Gain (avg)

89.2%

Best F1 (ADE Corpus v2)

In-depth Result Analysis

DDI-2013: Precision-Recall Trade-off

Recall 36.0% → 56.2% **but Precision** 27.6% → 24.7%

Model over-predicts relations due to fine-grained DDI sub-type confusion (MECHANISM / EFFECT / ADVISE / INT)

TAC-2017 / CADEC: Domain Transfer is Hard

TAC-2017: only **+1.2% F1** CADEC: only **+9.0% F1**

Patient-generated text (clinical notes, online forums) has informal and highly variable language — too different from training data

Key Insight

Fine-tuning does not teach the model WHAT biomedical relations are — the base model already knows. Fine-tuning teaches it WHEN to extract them. (Base Qwen3-8B already achieves 30% F1 on ADE/DDI in zero-shot mode)

Qualitative Error Analysis

Common failure modes of the fine-tuned model

Compound Entity Splitting

"trimethoprim-sulfamethoxazole" predicted as one entity,
but gold splits into two separate drugs.

Partial Entity Span

Predicted "Hypersensitivity" but gold is "Hypersensitivity reactions"
— model truncates multi-word adverse effects.

Relation Type Confusion

Predicted DDI-ADVISE + DDI-EFFECT instead of ADE relations.
Model struggles where DDI and ADE contexts overlap semantically.

Missing Secondary Relations

Predicted ADE(clozapine, TD) but missed ADE(clozapine, discomfort).
Model often captures only the primary relation in multi-relation sentences.

Conclusion & Future Outlook

Three-way validation: Rule-based (10%) → Zero-shot Qwen3-8B (22%) → rsLoRA Qwen3-8B (65%)

Each step up the hierarchy brings consistent, measurable improvement.

Fine-tuning provides the single largest gain.

- Augment training data with patient-generated text (TAC/CADEC domains)
- Entity normalization to reduce partial span errors
- Multi-relation supervision for complex sentences
- Combine rule-based system as post-processing filter to reduce false positives

DEMO

Q&A